AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956
SCHOOL OF COMPUTING CHENNAI

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

23CSE304 Embedded Systems Laboratory

Activity Sheet #10

Activity Name: Interrupt Programming

Name of Student:	**Agneay B Nair**
Registration No.:	**CH.SC.U4CSE24102**
Activity Date:	**21-08-26**
Submission Date:	**21-08-26**

Assessment Rubrics:

Component	Maximum Marks	Marks Allotted
Program	3	
Output	4	
Viva	2	
Timely Submission	1	
Total	**10**	

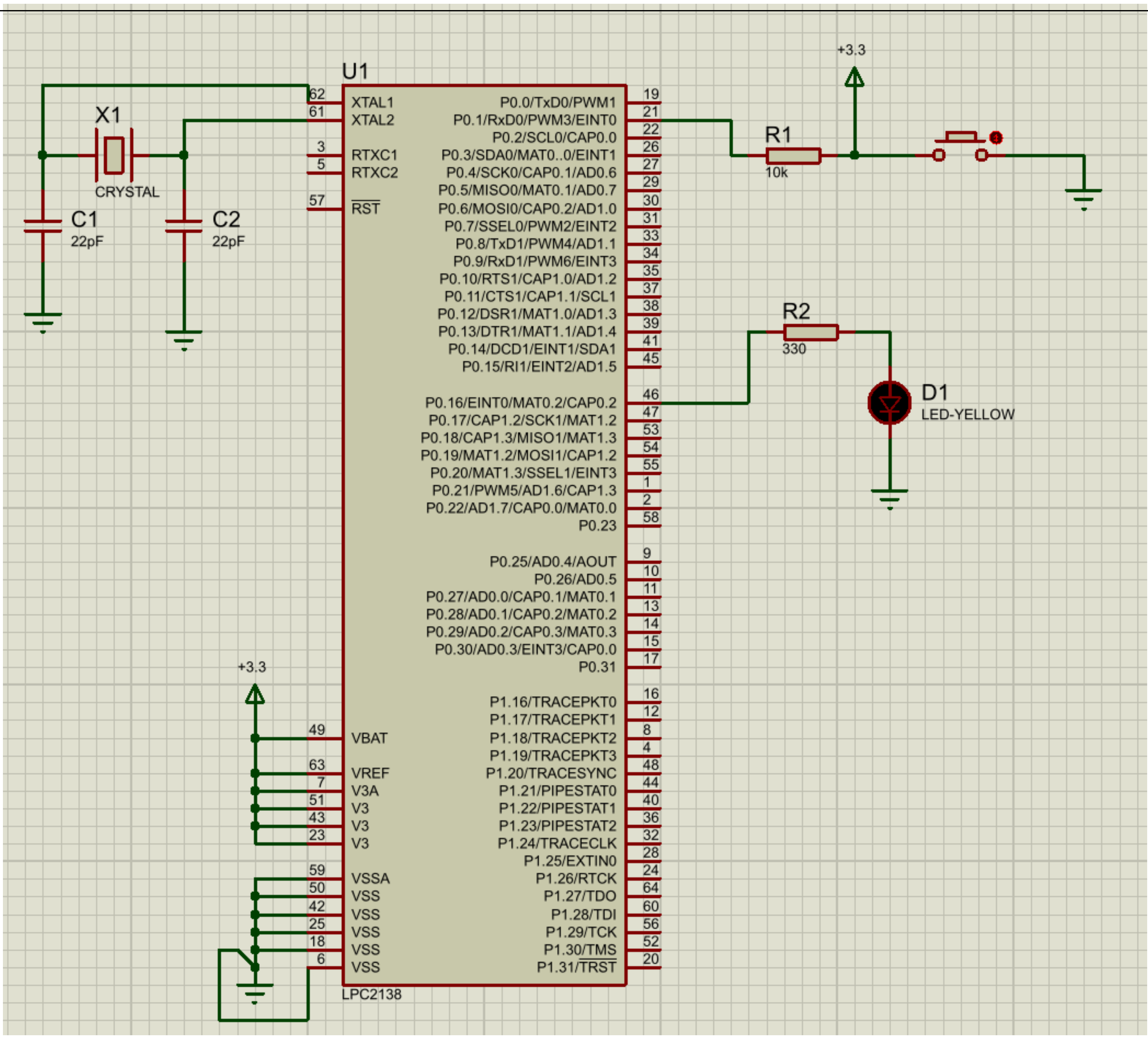
Faculty In-charge

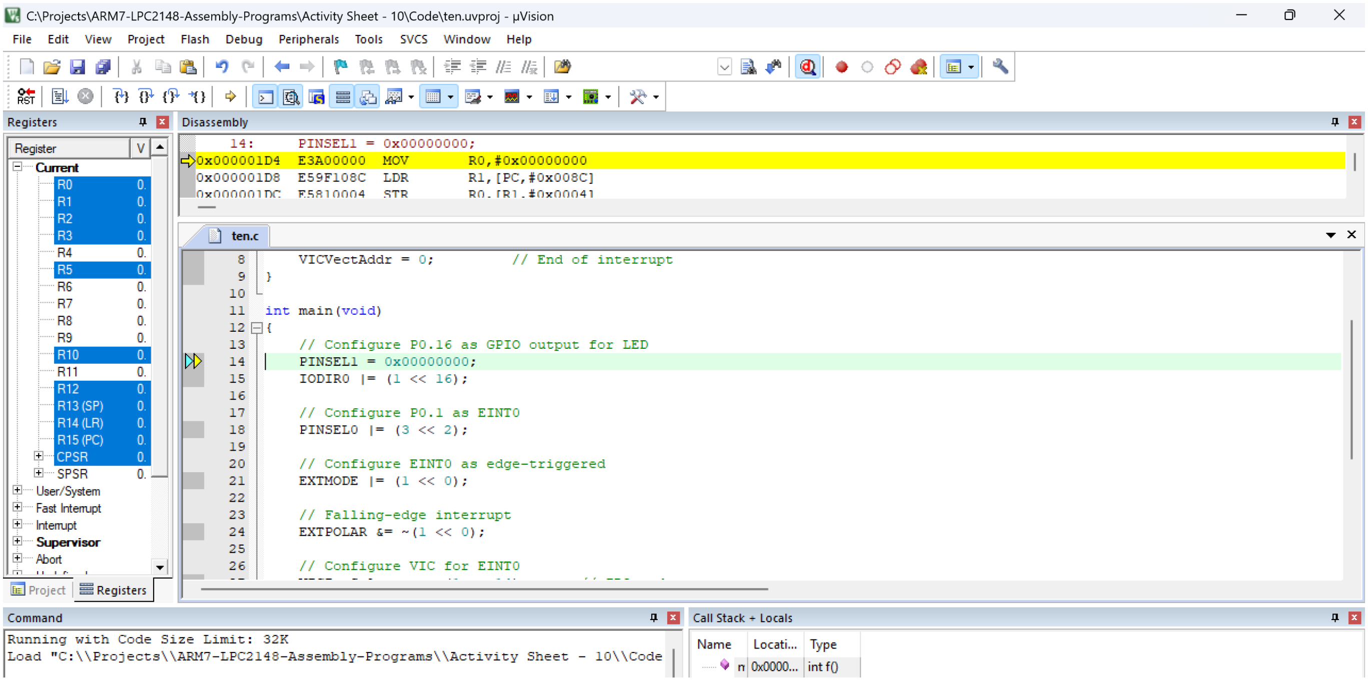
Write an embedded C program for ARM architecture to drive Interrupt module. Debug and verify the program using IDE on a development computer system.

CODE:

```
#include <LPC214x.h>

void EINT0_ISR(void) __irq
{
   IO0PIN ^= (1 << 16);     // Toggle LED

   EXTINT = (1 << 0);       // Clear EINT0 interrupt flag
   VICVectAddr = 0;         // End of interrupt
}

int main(void)
{
   // Configure P0.16 as GPIO output for LED
   PINSEL1 = 0x00000000;
   IODIR0 |= (1 << 16);

   // Configure P0.1 as EINT0
   PINSEL0 |= (3 << 2);

   // Configure EINT0 as edge-triggered
   EXTMODE |= (1 << 0);

   // Falling-edge interrupt
   EXTPOLAR &= ~(1 << 0);

   // Configure VIC for EINT0
   VICIntSelect &= ~(1 << 14);      // IRQ mode
   VICVectAddr0 = (unsigned long)EINT0_ISR;
   VICVectCntl0 = (1 << 5) | 14;    // Enable slot 0 for EINT0
   VICIntEnable = (1 << 14);        // Enable EINT0 interrupt

   // LED initially OFF
   IO0SET = (1 << 16);

   while(1)
   {
      // Main program does nothing
      // LED changes only when interrupt occurs
   }
}
```

OUTPUT:


U1
LPC2138
X1
CRYSTAL
C1
22pF
C2
22pF
XTAL1
XTAL2
RTXC1
RTXC2
RST
VBAT
VREF
V3A
V3
V3
V3
VSSA
VSS
VSS
VSS
VSS
VSS
P0.0/TxD0/PWM1
P0.1/RxD0/PWM3/EINT0
P0.2/SCL0/CAP0.0
P0.3/SDA0/MAT0..0/EINT1
P0.4/SCK0/CAP0.1/AD0.6
P0.5/MISO0/MAT0.1/AD0.7
P0.6/MOSI0/CAP0.2/AD1.0
P0.7/SSEL0/PWM2/EINT2
P0.8/TxD1/PWM4/AD1.1
P0.9/RxD1/PWM6/EINT3
P0.10/RTS1/CAP1.0/AD1.2
P0.11/CTS1/CAP1.1/SCL1
P0.12/DSR1/MAT1.0/AD1.3
P0.13/DTR1/MAT1.1/AD1.4
P0.14/DCD1/EINT1/SDA1
P0.15/RI1/EINT2/AD1.5
P0.16/EINT0/MAT0.2/CAP0.2
P0.17/CAP1.2/SCK1/MAT1.2
P0.18/CAP1.3/MISO1/MAT1.3
P0.19/MAT1.2/MOSI1/CAP1.2
P0.20/MAT1.3/SSEL1/EINT3
P0.21/PWM5/AD1.6/CAP1.3
P0.22/AD1.7/CAP0.0/MAT0.0
P0.23
P0.25/AD0.4/AOUT
P0.26/AD0.5
P0.27/AD0.0/CAP0.1/MAT0.1
P0.28/AD0.1/CAP0.2/MAT0.2
P0.29/AD0.2/CAP0.3/MAT0.3
P0.30/AD0.3/EINT3/CAP0.0
P0.31
P1.16/TRACEPKT0
P1.17/TRACEPKT1
P1.18/TRACEPKT2
P1.19/TRACEPKT3
P1.20/TRACESYNC
P1.21/PIPESTAT0
P1.22/PIPESTAT1
P1.23/PIPESTAT2
P1.24/TRACECLK
P1.25/EXTIN0
P1.26/RTCK
P1.27/TDO
P1.28/TDI
P1.29/TCK
P1.30/TMS
P1.31/TRST
+3.3
R1
10k
R2
330
D1
LED-YELLOW


C:\Projects\ARM7-LPC2148-Assembly-Programs\Activity Sheet - 10\Code\ten.uvproj - µVision
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Disassembly
Register
Current
R0 R1 R2 R3 R4 R5 R6 R7 R8 R9 R10 R11 R12 R13 (SP) R14 (LR) R15 (PC) CPSR SPSR
User/System
Fast Interrupt
Interrupt
Supervisor
Abort
Project
Registers
14: PINSEL1 = 0x00000000;
0x000001D4 E3A00000 MOV R0,#0x00000000
0x000001D8 E59F108C LDR R1,[PC,#0x008C]
ten.c
VICVectAddr = 0; // End of interrupt
}
int main(void)
{
// Configure P0.16 as GPIO output for LED
PINSEL1 = 0x00000000;
IODIR0 |= (1 << 16);
// Configure P0.1 as EINT0
PINSEL0 |= (3 << 2);
// Configure EINT0 as edge-triggered
EXTMODE |= (1 << 0);
// Falling-edge interrupt
EXTPOLAR &= ~(1 << 0);
// Configure VIC for EINT0
Command
Running with Code Size Limit: 32K
Load "C:\\Projects\\ARM7-LPC2148-Assembly-Programs\\Activity Sheet - 10\\Code
Call Stack + Locals
Name
Locati...
Type
int f()